

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 3**

**COURSE CODE:** CSA5110 **COURSE NAME:** Cryptography and Network Security

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### **COURSE OUTCOME COVERED**

**CO4:** *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

*Assessment Tool Weightage: 10%*

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**QUESTIONS: 5**

**TOTAL MARKS: 25**

### **Annexure A**

#### ***Comparative Analysis***

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#### ***Code of Conduct:***

*I SAIRAM D 192424111 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: SAIRAM D***

## **Annexure A**

### **Comparative Analysis**

1 . Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec with respect to security services, performance overhead, and suitability for different network scenarios. Analyze which is more appropriate for a VPN deployment.

#### **SOLUTION:**

##### Introduction

IPSec is a network security framework used to protect IP communication over untrusted networks such as the public Internet. Two important IPSec protocols are Authentication Header (AH) and Encapsulating Security Payload (ESP).

AH mainly provides authentication, data integrity, and anti-replay protection, whereas ESP can provide confidentiality, authentication, integrity, and anti-replay protection. The selection of AH or ESP depends on the security requirements, network environment, and performance requirements.

#### 1. Authentication Header (AH)

Authentication Header provides authentication, data integrity, and anti-replay protection. It verifies that the packet came from an authenticated source and that the packet was not modified during transmission.

AH does not provide confidentiality because it does not encrypt the actual data. Therefore, the contents of the communication remain visible to an attacker monitoring the network.

The basic AH packet structure is:

IP Header | AH Header | Original Data

The main advantages of AH are that it provides strong integrity protection, authenticates the source of the packet, and provides protection against replay attacks. AH also has lower processing overhead compared with ESP when encryption is enabled.

The main disadvantages are that AH does not provide confidentiality and the actual application data remains visible. AH can also have compatibility problems with Network Address Translation (NAT) because AH protects parts of the IP header that may be modified by NAT devices. Due to these limitations, AH is rarely used in modern VPN deployments.

## 2. Encapsulating Security Payload (ESP)

Encapsulating Security Payload is the more commonly used IPSec protocol. ESP can provide confidentiality through encryption and can also provide integrity, authentication, and anti-replay protection.

A simplified ESP packet structure is:

IP Header | ESP Header | Encrypted Data | ESP Authentication Data

ESP encrypts the protected data, preventing unauthorized users from reading the communication. For example, if a branch office sends confidential financial information through the public Internet, ESP encryption prevents attackers from understanding the captured traffic.

The main advantages of ESP are confidentiality through encryption, data integrity, authentication, and anti-replay protection. ESP is widely used in modern VPN technologies, especially for site-to-site VPNs using Tunnel Mode.

The main disadvantage of ESP is that encryption and decryption require additional processing. It also introduces some packet overhead. However, modern VPN gateways often provide hardware cryptographic acceleration, which reduces the performance impact.

## 3. Security Services Comparison

AH provides authentication, integrity, and anti-replay protection. ESP can provide confidentiality, authentication, integrity, and anti-replay protection.

The major difference between AH and ESP is confidentiality.

AH does not encrypt the data, whereas ESP can encrypt the data.

For example, if a company sends confidential information such as:

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using AH, the attacker may still be able to see the actual information even though the packet is protected against modification.

When ESP encryption is used, the attacker sees encrypted information instead of the original data. Therefore, ESP provides better protection for confidential business communication.

## 4. Performance Overhead

AH generally has lower processing overhead because it does not perform encryption. Its main operations are authentication and integrity verification.

ESP normally has higher processing requirements because it performs encryption and decryption along with integrity and authentication.

AH provides lower processing overhead.

ESP provides higher processing overhead but significantly stronger security.

The performance impact of ESP can be reduced by using modern VPN gateways, hardware cryptographic acceleration, efficient encryption algorithms, and proper network configuration.

Therefore, although AH may provide better performance in some situations, ESP provides much stronger overall security.

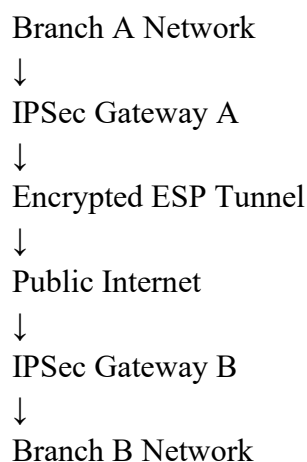
## 5. Suitability for Different Network Scenarios

AH can be considered for host-to-host communication when authentication and integrity are required but confidentiality is not necessary.

ESP is more suitable when communication contains sensitive or confidential information.

For site-to-site VPNs, ESP is strongly preferred because it can protect the complete original IP packet using Tunnel Mode.

The communication can be represented as:



For remote-access VPNs, ESP is also more appropriate because remote users may transmit passwords, company files, database information, and other confidential data through the Internet.

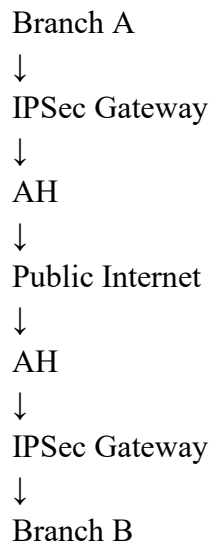
AH can be considered for non-confidential traffic where only integrity and authentication are required. However, ESP can also provide integrity protection without requiring encryption in configurations where confidentiality is not needed.

## 6. VPN Deployment Analysis

Consider a multinational organization with branch offices communicating through the public Internet. The branches exchange financial information, employee information, customer information, database communication, internal application data, and file transfers.

Using AH alone would not be sufficient because the data would remain visible to attackers.

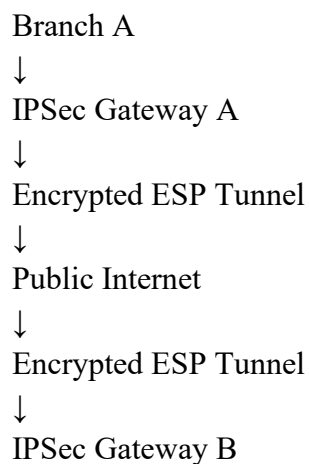
With AH, the communication can be represented as:



The communication is authenticated and protected from modification, but the actual data is not encrypted.

An attacker monitoring the Internet connection may still be able to observe the contents of the communication.

Using ESP Tunnel Mode provides a stronger solution:





Branch B

The original IP packet is encapsulated and encrypted. Therefore, an attacker monitoring the Internet connection cannot easily read the internal communication.

## 7. Why ESP is More Appropriate for VPN Deployment

ESP is more appropriate for VPN deployment because a VPN is designed to create a secure communication channel over an untrusted network.

A secure VPN generally requires confidentiality, integrity, authentication, and anti-replay protection.

ESP can provide all these security services.

AH cannot provide confidentiality because it does not encrypt the data. Therefore, AH alone is not sufficient when the VPN carries sensitive organizational information.

ESP Tunnel Mode is particularly suitable for site-to-site VPNs because it protects the complete original IP packet and hides the internal network communication from external users.

## 8. Performance versus Security Analysis

AH provides lower processing overhead because it does not encrypt the data. However, its security capability is limited because it does not provide confidentiality.

ESP requires additional processing for encryption and decryption. This creates more overhead compared with AH. However, ESP provides much stronger protection and is suitable for confidential enterprise communication.

In a business environment, the additional processing cost of ESP is generally acceptable because protecting confidential information is more important than achieving a small reduction in processing overhead.

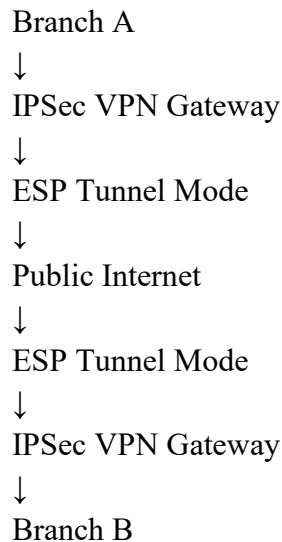
Modern VPN hardware can also use hardware acceleration to improve ESP performance.

Therefore, the enterprise should prefer security over the small performance advantage provided by AH.

## 9. Final Recommendation

For a modern enterprise VPN deployment, ESP should be selected instead of AH.

The recommended configuration is:



ESP should be configured with strong encryption, integrity protection, authentication, and anti-replay protection.

AH can still be considered in specialized situations where confidentiality is not required and only authentication and integrity are needed.

Using both AH and ESP together is technically possible, but it is generally unnecessary for normal enterprise VPN deployment because ESP can already provide the required security services.

## Conclusion

AH and ESP are important IPSec protocols that provide different levels of protection. AH provides authentication, integrity, and anti-replay protection but does not provide confidentiality. ESP provides confidentiality along with authentication, integrity, and anti-replay protection.

AH has lower processing overhead because it does not perform encryption. ESP has higher processing overhead because it performs encryption and decryption, but modern hardware and cryptographic acceleration can reduce this impact.

For host-to-host communication where only authentication and integrity are required, AH may be considered. However, for enterprise VPNs, remote-access VPNs, and site-to-site VPNs carrying confidential information, ESP is the better choice.

Therefore, ESP in Tunnel Mode is the most appropriate solution for a modern enterprise VPN deployment because it provides confidentiality, integrity, authentication, and anti-replay protection while securely transmitting organizational data over the public Internet.

## RUBRICS FOR CALCULATION:

| Criteria                | Weightage (%) | Level 4 (Exemplary)                                                           | Level 3 (Proficient)                            | Level 2 (Developing)                       | Level 1 (Beginning)                      |
|-------------------------|---------------|-------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------|------------------------------------------|
| Scope of Items          | 20%           | Selects highly relevant items with clear scope and justification              | Selects appropriate items with minor gaps       | Selection is limited or partially relevant | Items are unclear or not relevant        |
| Comparison Depth        | 25%           | Provides detailed and comprehensive comparison across multiple aspects        | Provides adequate comparison with some depth    | Limited comparison with few aspects        | Very minimal or no comparison            |
| Use of Evidence         | 20%           | Supports comparison with accurate data, examples, or references               | Uses some supporting evidence with minor gaps   | Limited or weak supporting evidence        | No supporting evidence provided          |
| Critical Thinking       | 20%           | Demonstrates strong critical thinking with clear interpretation and reasoning | Shows reasonable interpretation with minor gaps | Basic interpretation; lacks depth          | No meaningful interpretation             |
| Clarity of Presentation | 15%           | Well-organized, clear, and logically structured presentation                  | Generally clear with minor issues               | Somewhat unclear or poorly structured      | Disorganized and difficult to understand |